

# PIGSTI

## 1 Preprocessing

### Raw reads

(FASTQ)

## 2 Host route

### Trim adapters

(Adapter Removal)

### Host identification

(FastQ Screen)

### Map host

(BWA / Bowtie2)

### Map mtDNA

(BWA / Bowtie2)

## 3 Metagenomics screening

### Dedup

(PRINSEQ++)

### Mammals removal

(Bowtie2)

### Pool samples

(cat · pigz)

### Metagenome screening

(KrakenUniq)

### Metagenome screening

(MALT)

### Merge samples

(samtools)

### Soft clipping

(Python)

### Damage

(DamageProfiler)

#### Host species



## 4 Pathogen route

### Pathogen detection

(E-value · MaltExtract)

### Pathogen mapping

(BWA / Bowtie2)

### Authentication

(composite score)

### Authentication details

ANI

Entropy

Breadth

5' C→T damage

Edit distance

Mapping ratio

Genus rank

### Exportables

(Excel · PDF)

### Genetic sexing

(sexing\_residual)

- Preprocessing
- Metagenomics
- Mapping
- Optional
- Host
- Detection
- Authentication